AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS 2021

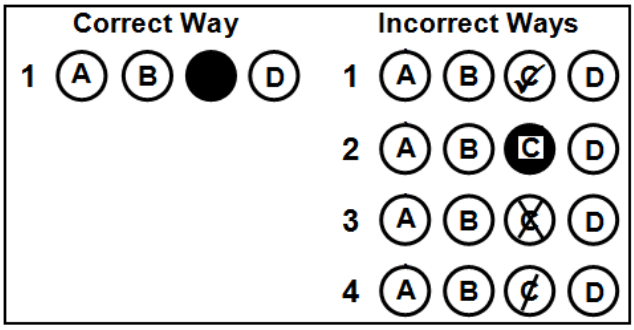
Business Statistics

Time: 1 hour 20 minutes Marks: 40

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 40 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. The marks obtained on the 40 MCQs will be equated to the total marks of 50 for the theory examination results.
8. You may use a scientific calculator if you wish.

1. Which of the given statements is FALSE for descriptive statistics?

   A. It summarises the sample data.
   B. It presents data in meaningful way.
   C. It draws conclusion about population.
   D. It represents data through graphs and tables.

2. An example of quantitative variable is a person's

   A. nationality.
   B. bank balance.
   C. preferred airline.
   D. favourite author.

3. Which of the following is NOT a qualitative data?

   A. Nature of work
   B. Colour of human eyes
   C. Brand of a washing machine
   D. Number of homes in an area

Use the given information to answer Q.4, Q.5 and Q.6.

The table gives information about the number of employees in a certain organisation and the distance from their home to the office.

Distance (km)	Number of Employees
1 – 4	15
5 – 8	22
9 – 12	45
13 – 16	23
17 – 20	10

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4. The class width for the given data is

   A. 3
   B. 4
   C. 5
   D. 19

5. The class boundary associated to the highest frequency for the given data is

   A. 8.5 – 11.5
   B. 9.5 – 12.5
   C. 8.5 – 12.5
   D. 9.5 – 11.5

6. The class mark of the last class is

A. 1.5
B. 16.5
C. 18.5
D. 20.5

7. The given table shows the number of screws and their lengths in inches.

Length of Screws (inches)	Number of Screws	Cumulative Frequency
1 – 3	5	5
4 – 6	7	12
7 – 9	4	16
10 – 12	*m*	?

If the cumulative frequency of the screws having lengths between 10 to12 inches is 22, then the value of *m*

A. is 6
B. is 16
C. is 22
D. cannot be determined

Use the given information to answer Q.8 and Q.9.

Items	Frequency
I	*a*
II	*a*
III	*a*
IV	*a*

8. The relative frequency of item III

A. is 25%
B. is 75%
C. cannot be determined
D. is between 25% and 75%

9. If the cumulative frequency of item IV is 60, then the value of 3*a*

A. is 15
B. is 20
C. is 45
D. cannot be determined

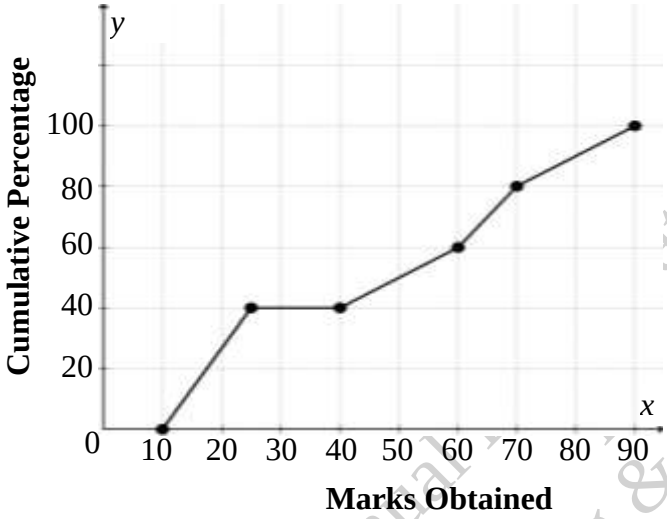
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10. Which of the following data can be represented in a pie chart?

I. Time
II. Gender
III. Hair colour

A. I only
B. III only
C. I and II
D. II and III

11. The given graph shows the marks obtained by a class in Mathematics.

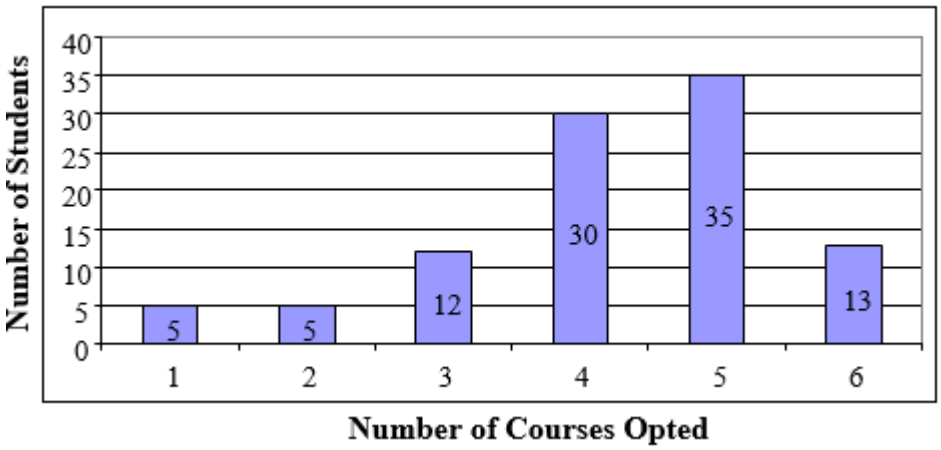


The percentage of candidates who obtained marks between 70 to 90 is

A. 20%
B. 40%
C. 80%
D. 100%

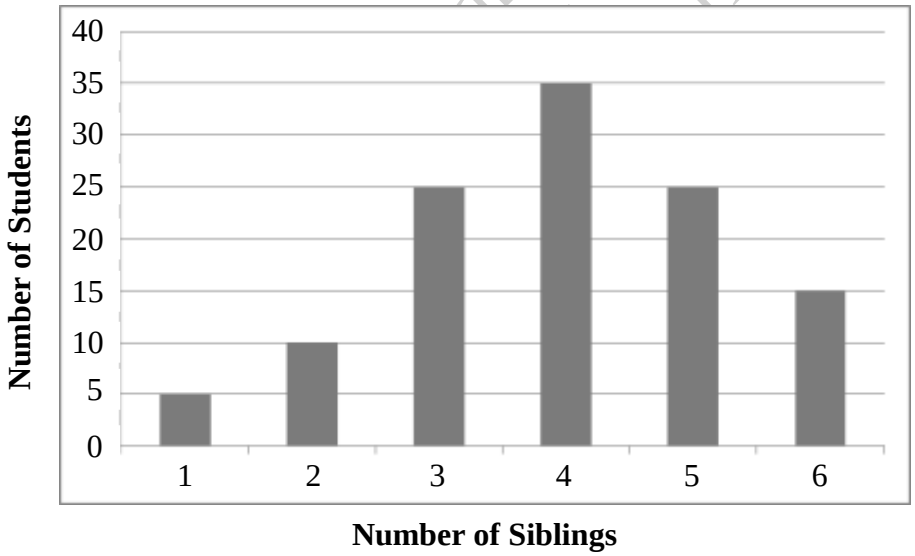
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12. The graph represents data about the number of courses opted by 100 students registered in an MBA class. The number of students who opted for fewer than 3 courses is



A. 78
B. 22
C. 12
D. 10

13. The given bar graph represents the number of students and their siblings in a school.

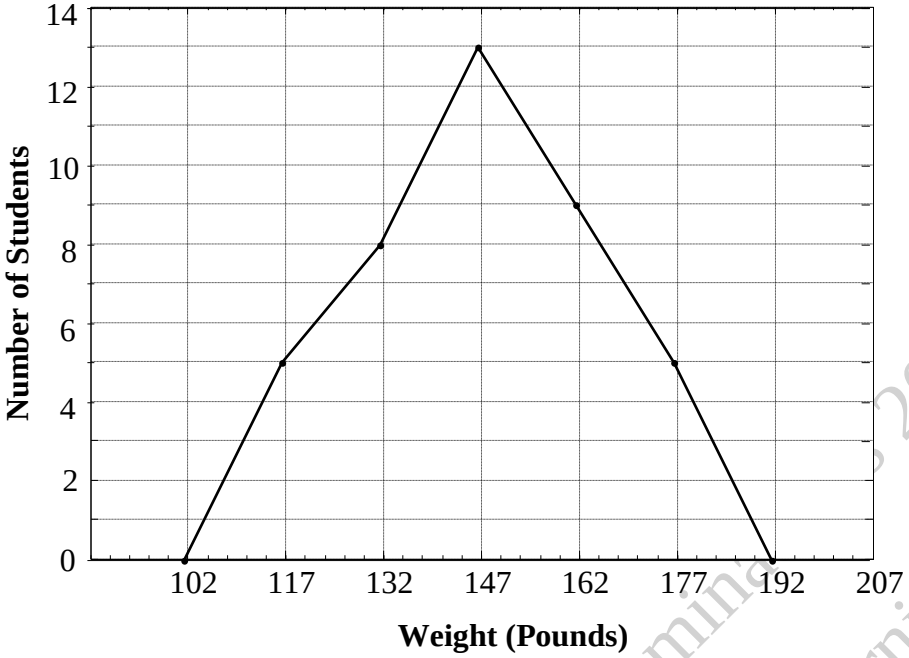


The number of students having more than four siblings is

A. 35
B. 40
C. 75
D. 100

Use the given information to answer Q.14 and Q.15.

The weight (in pounds) of students of a class is represented by the given frequency polygon.



14. The total number of students in this class is

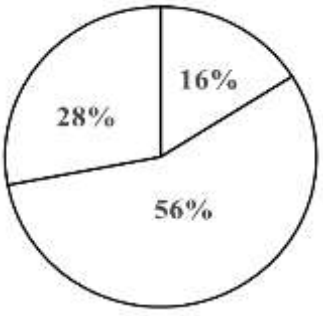
A. 13
B. 30
C. 40
D. 42

15. The percentage of students that have weight less than or equal to 147 pound is

A. 32.5%
B. 35%
C. 53%
D. 65%

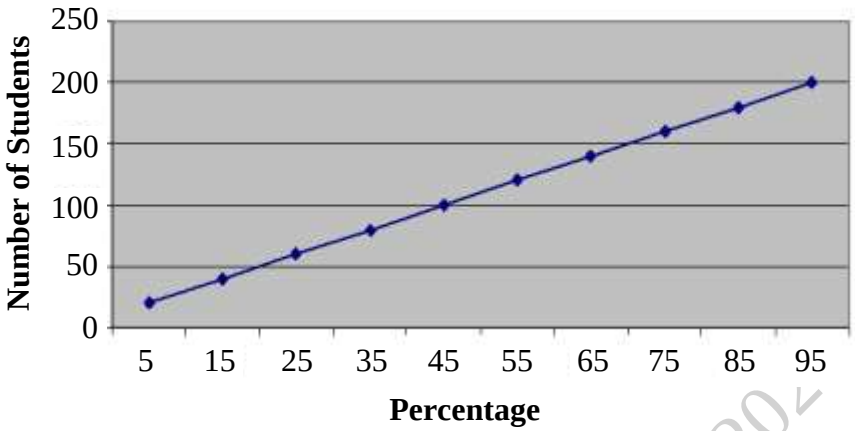
16. In the given pie chart, the angle corresponding to the largest sector is

A. 56°
B. 100.8^o
C. 201.6^o
D. 270°



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17. In the given cumulative frequency curve, the approximate number of students who achieved more than 65% of marks is



A. 60
B. 140
C. 150
D. 200

18. The mode of the data 1, 3, 9, 3, 1, 1, 8 and 8 is equal to

A. 1
B. $\frac{1}{2}$
C. 2
D. 6

19. If $Y - a = X$ and $\overline{X} + a = 14$, then the value of $2\overline{Y}$

A. is 7
B. is 14
C. is 28
D. cannot be determined

20. If the arithmetic mean of 15, 20, *a*, 35 and 25 is 23, then the value of *a* will be

A. 20
B. 23
C. 25
D. 30

21. If the mean of variable *X* is 150, then the mean of variable $Y = \frac{2X}{3} - 5$ will be

A. 95
B. 100
C. 220
D. 225

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22. If the median of the values 2, 5, 7, x, $2x$, $2x + 2$, 25, 50 is 13.5, then the value of x will be

A. 7
B. 9
C. 18
D. 24

23. The empirical relation between mean, median and mode is given as

mean – mode = 3 (mean – median).

For the data 1, m, 5, 7, 13, 1, 7 and 7, where m is greater than 13, the value of the expression

3 medians – 2 mean is

A. 1
B. 4
C. 7
D. 10

24. If the mode of X is 3, then the mode of $2X + 4$ will be

A. 3
B. 6
C. 7
D. 10

25. The median of 3, 6, 10, a, 5 and 8, when $a < 3$, is equal to

A. $\frac{a+10}{2}$
B. $\frac{11}{2}$
C. 11
D. 7

26. If $Y - k = X$ and $\overline{X} + k = 20$, then the value of $\overline{Y} - 5$ is

A. 5
B. 10
C. 15
D. 20

27. The variance of the data 4, 4, 4, 4, 4, 4, 4, 4 is

A. ± 2
B. 0
C. 4
D. 16

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28. If the variance of X is 25, then the variance of $\frac{4}{5}X - 5$ will be

A. 11
B. 15
C. 16
D. 20

29. For a data set mean is 2, $\sum(X-2)^2 = 250$ and $\sigma = 5$, then the value of N is equal to

A. 5
B. 10
C. 25
D. 50

30. What would be the value of p if the range of the given data is 14.5?

Class Interval	$-2.5 < x < 0$	$0.5 < x < 3$	$3.5 < x < 6$	$6.5 < x < 9$	$9.5 < x < p$
Frequency	2	?	5	8	?

A. 12
B. 12.5
C. 14.5
D. 15

31. If the standard deviation of 1, 2, 3, 4, 5 is 1.4142, then the variance of 2, 4, 6, 8, 10 is equal to

A. 1.4142
B. 2.8282
C. 4.0000
D. 8.0000

32. Which one of the following options is TRUE for standard deviation?

I. It has no units.
II. It is positive.
III. It is square root of variance.

A. I only
B. II only
C. I and III
D. II and III

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33. The events E and F are mutually exclusive. If $P(E \text{ and } F) = m$, then the value of m is

A. 0
B. $\frac{1}{4}$
C. $\frac{1}{2}$
D. 1

34. A person can select one magazine from 8 different magazines, one newspaper from 10 different newspapers and one story book from 12 different story books. The possible ways in which the person can make the selection is

A. 28
B. 30
C. 792
D. 960

35. If A and B are two complementary events, then

A. $P(A) + P(B) = 0$
B. $P(A) + P(B) = 1$
C. $P(A) \times P(B) = 1$
D. $P(A) \times P(B) = 0$

36. The number of ways in which 8 distinct letters chosen from 26 letters of English alphabets are

A. 26^8
B. $\frac{26!}{18!}$
C. $\frac{26!}{8!}$
D. $\frac{26!}{8!8!}$

37. In a stack of nine books, six books are of science fiction and three books are of mysteries. The different number of ways to select three books such that two books are of science fictions and one book of mysteries is

A. $\frac{9!}{6!3!}$
B. $6^2 \times 3$
C. $\frac{6!}{4!} \times 3!$
D. $\frac{6!}{2!4!} \times \frac{3!}{2!}$

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38. The events M and N are independent. If $P(M/N)=1-x$ and $P(M)=\frac{3}{5}$, then the value of x is

A. 1

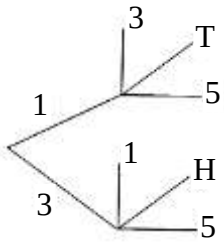
B. $\frac{2}{5}$

C. $\frac{3}{5}$

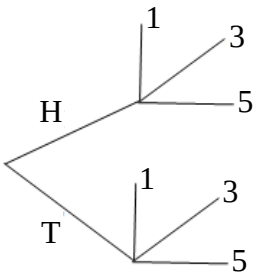
D. $\frac{8}{5}$

39. A spinner having number 1, 3, 5 and a coin are thrown together. The tree diagram which illustrates the possible outcomes of these two will be

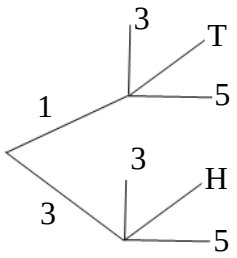
A.



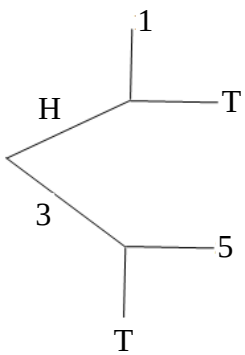
B.



C.



D.



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40. The number of possible permutations of the word BLACKBOARD is

A. 10
B. 10!
C. $\frac{10!}{4!}$
D. $\frac{10!}{2! \times 2!}$

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